



A Multifaceted Reasoning Network for Explainable Fake News Detection

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ABSTRACT

Fake news detection involves developing techniques to identify and flag misleading or false information disseminated through media sources. Current efforts often use limited information for categorization, lacking comprehensive data integration and explanation of results. Additionally, the substantial noise generated by multi-source data presents extra challenges to fake news detection. To address these problems, we propose a novel **Multifaceted Reasoning Network for Explainable Fake News Detection** (MRE-FND). This model constructs two heterogeneous graphs to learn about social network information and news content knowledge, including news content, social networks, knowledge graphs, and external news data. Utilizing graph information bottleneck theory, it eliminates noise from multifaceted data and extracts key information for fake news detection. An interpretable reasoning module is designed to provide clear explanations for the classification results. Our proposition undergoes extensive evaluation on three popular datasets, Politifact, Gossipcop and PHEME, which consist of 495, 15707 and 2189 news, respectively. Our model achieved state-of-the-art results across all metrics on three datasets. Specifically, our model achieved accuracy rates of 92.9%, 83.4% and 84.7% on the Politifact, Gossipcop and PHEME datasets, respectively, demonstrating improvements of 2.0, 0.8 and 1.1 percentage points over the baseline, thus establishing the superiority of our model. Further analysis indicates that our model can effectively handle redundant information in multi-faceted data, enhancing the performance of fake news detection while also providing multifaceted explanations for the classification results.

1. Introduction

The proliferation of social media platforms has significantly streamlined the accessibility and cost efficiency of news dissemination (Khattar, Goud, Gupta, & Varma, 2019). Additionally, it enables news to be rapidly disseminated in a multimodal format. In contrast to conventional textual news formats, multimodal news compositions incorporating images or videos offer heightened and more engaging reading experiences (Wang et al., 2018). However, the inclusion of multimodal elements renders such content susceptible to malicious manipulations, such as the misrepresentation or fabrication of images, consequently accelerating the proliferation of misinformation (Khattar et al., 2019). Hence, there is an urgent need for the automatic assessment of news credibility to detect instances of fake news therein, thereby mitigating or preventing the risks associated with its dissemination.

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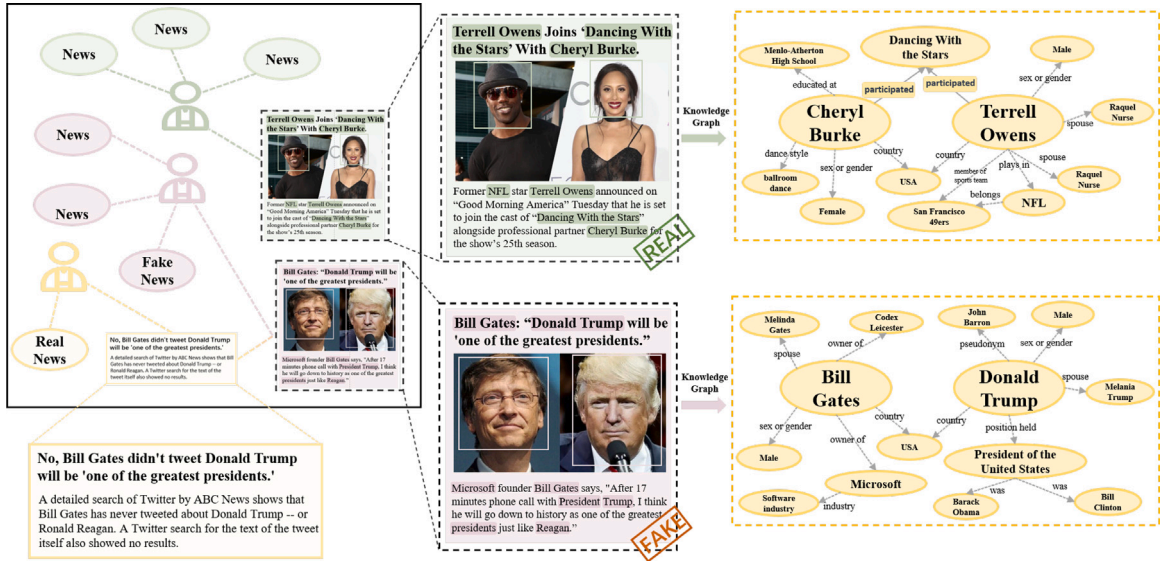


Fig. 1. Illustrating through examples, it is insufficient to rely solely on partial external information for the complete discernment of the authenticity of news.

The initial methods to detect fake news are based on the textual content of news articles (Castillo, Mendoza, & Poblete, 2011; Chen, Li, Yin, & Zhang, 2018; Huh, Liu, Owens, & Efron, 2018; Luembe, Li, Li, Liu, & Xu, 2023; Przybyla, 2020; Qi, Cao, Yang, Guo, & Li, 2019; Wu, Yang, & Zhu, 2015; Zhang et al., 2021), involving the classification through the exploration of text features such as sentiment, keywords, and thematic information. For instance, Castillo et al. (2011) utilize linguistic features including special characters, sentiment (positive/negative) words, emoticons, and others to identify fake news. Subsequently, multimodal fake news detection methods (Jin, Cao, Guo, Zhang, & Luo, 2017; Jing, Wu, Sun, Fang, & Zhang, 2023; Khattar et al., 2019; Qi et al., 2021; Wu, Zhan, Zhang, Wang, & Xu, 2021; Zhang, Zhang, Zhu, Liu, & Xu, 2022) are introduced, aiming to enhance the accuracy of fake news detection by leveraging both visual and textual features. Li et al. (2021) propose a novel paradigm for detecting false information by aligning and fusing multimodal entities. In order to evade detection by advanced detection models, recent instances of fake news typically exhibit highly sophisticated fabrication, rendering them difficult to discern based solely on their textual content. As a result, some efforts begin to incorporate external knowledge (Dou, Shu, Xia, Yu, & Sun, 2021; Dou et al., 2021, 2021, 2021; Ren & Zhang, 2021, 2021; Sheng et al., 2022, 2022, 2022, 2022; Su, Yang, Wu, & Zhang, 2023; Su et al., 2023) to assist in fake news detection.

Existing knowledge-enhanced detection methods can be classified into two categories. The first category involves referencing social network information (Bazmi, Asadpour, & Shakery, 2023; Dou et al., 2021, 2021; Fang et al., 2024; Peng, Jian, Kan, Qiao, & Li, 2024; Ren & Zhang, 2021; Sheng et al., 2022, 2022; Su et al., 2023). It is indicated that trustworthy users are more likely to share credible news, while untrustworthy users are more likely to disseminate misinformation (Su et al., 2023). Consequently, some works leverage the construction of user propagation networks to assist in fake news detection (Dou et al., 2021; Ren & Zhang, 2021; Sheng et al., 2022; Su et al., 2023). Ren and Zhang (2021) construct a heterogeneous information network based on information content, publishing users, and information topics, and utilized hierarchical attention mechanisms to learn node representations. In addition, some works reference external news information or user comments to assist in fake news detection (Dou et al., 2021; Sheng et al., 2022). The recent opinions of mainstream media and public attention serve as crucial sources of inspiration for misinformation creation, while external news and user comments can additionally enrich textual features. The second category involves leveraging knowledge graphs by associating entities mentioned in the news with the corresponding entities in the knowledge graph (Pan et al., 2018; Sun et al., 2023; Wang, Qian, Hu, Fang, & Xu, 2020). The learned entity embeddings are then incorporated as supplementary information to complement fake news detection. Hu et al. (2021) propose a novel model called CompareNet, which directly compares information with external knowledge for fake news detection. In a nutshell, external common knowledge and multimodal news content are indispensable for detecting fake news successfully. The first category of methods focuses more on exploring relationships between news articles, which we summarize as coarse-grained features. In contrast, the second category of methods concentrates on mining fine-grained entity knowledge within news articles, which we summarize as fine-grained features. However, these methods still exhibit the following limitations.

Firstly, with technological advancements, recent instances of fake news are often characterized by a heightened level of meticulous fabrication. More than relying on partial external information is required to discern the authenticity of news completely. For example, in the first instance depicted in Fig. 1, relying solely on social networks is insufficient to determine the authenticity of the news "Terrell Owens Joins 'Dancing With the Stars' With Cheryl Burke." It necessitates the introduction of a knowledge graph to seek criteria for judgment. Within the knowledge graph, both Cheryl Burke and Terrell Owens are documented participants in the "Dancing With the Stars" program. Hence, the authenticity of the news is confirmed as real. In the second instance illustrated

in Fig. 1, only the news content and knowledge graph are insufficient to identify the reasons for the inconsistency between the news “Bill Gates: ‘Donald Trump will be one of the greatest presidents.’” and the factual reality. From social networks, it can be observed that there are related news articles debunking this claim. Additionally, the users who disseminated this news may have a history of publishing substantial misinformation. Consequently, it can be deduced that the news in question is fake. However, existing methods are not effective in integrating knowledge from multiple sources and different granularities.

Secondly, existing methods fail to select valuable external knowledge accurately. As depicted in Fig. 1, a significant amount of irrelevant noise exists within both social networks and knowledge graphs. Prior methods leveraging knowledge graphs typically employ entity linking to incorporate external knowledge without the ability to filter information within the knowledge graph. Methods utilizing social networks heavily rely on the quality of manually constructed news propagation networks. The presence of irrelevant noise in multisource information significantly impacts the performance of fake news detection.

Thirdly, existing fake news detection methods lack explanatory power regarding classification outcomes. Most efforts focus on detecting fake news using latent features without explaining the “reasons” behind a news article being classified as fake. Only a few studies (Lu & Li, 2020; Shu, Cui, Wang, Lee, & Liu, 2019; Yang et al., 2022) focusing on explainable fake news detection typically capture only some key sentences in news articles or key nodes in propagation networks. The explanations generated by these methods are rather one-sided, as they fail to provide explanations for news classification results from both the coarse-grained and fine-grained perspectives. Since the correlations between the multimodal information of the news are hierarchical and heterogeneous, the urgent task is to design an explainable fake news detection model capable of exploring the hierarchical relationships between coarse-grained social network and fine-grained knowledge graph.

To tackle the limitations above, our motivation lies in extracting effective information of varying granularities for fake news detection and providing distinct explanations tailored to different granularities of information. However, this is a challenging task. Firstly, each news article on social media is accompanied by a plethora of relevant users, related news, and associated knowledge, making it extremely challenging to extract useful information from this extensive pool of data. Secondly, users, news text, news images, and the knowledge graph are heterogeneous, making it challenging to construct a unified framework that can integrate various information modalities. Thirdly, providing diverse explanations for fake news detection classification results based on knowledge from different sources is challenging.

To address these problems, a novel **Multifaceted Reasoning Network for Explainable Fake News Detection** (MRE-FND) is proposed. This model integrates multi-faceted information, such as news content, social networks, knowledge graphs, and external news data, to detect fake news. It also explains classification results by learning the hierarchical and heterogeneous correlation between the multimodal information. Specifically, two heterogeneous graphs are constructed to represent the coarse-grained social network features and the fine-grained news knowledge, respectively, facilitating hierarchical modeling of new features. For training large-scale graphs while reducing noise, distinct graph sampling methods are devised for the coarse-grained and fine-grained graphs. To optimize the manually constructed graph structure, a Dual Graph Neural Network model (DGNN) is proposed to eliminate redundant edges or nodes and construct new edges. Moreover, an Explainable Reasoning Module (ERM) is developed to output key nodes that support the classification, providing explicit explanations for the model’s classification results. Lastly, a multi-task joint optimization framework is designed to optimize the model. Our proposition undergoes extensive evaluation on three popular datasets, and the experimental results validate the superiority of our proposal. Our significant contributions are summarized as follows:

- In contrast to previous fake news detection methods, we integrate multi-source and multi-granularity information, including news content, social networks, external news, news topics, and knowledge graphs, into our approach to fake news detection. Furthermore, explanations for fake news classification are provided across various granularities.
- A novel Multifaceted Reasoning Network for Explainable Fake News Detection is proposed, in which a dual graph neural network model (DGNN) is introduced to eliminate redundant edges or nodes and construct new edges. An explainable reasoning module (ERM) is proposed to output key nodes that support classification.
- Extensive experiments on real-world datasets reveal that our proposal consistently outperforms SOTA fake news detection baselines.

2. Research objectives

In order to establish an explainable fake news detection framework that encompasses users, source news, related news, and knowledge graphs, we set three research objectives:

- **RO1:** To construct a dual-layer heterogeneous graph model encompassing multiple sources of data, including users, source news, related news, and knowledge graphs (Sections 5.1 and 5.2).
- **RO2:** To design a dual graph neural network to extract and integrate valuable information from multiple sources (Section 5.3).
- **RO3:** To design a reasoning network to extract meaningful knowledge from intricate information and generate classification explanations based on it (Section 5.4).

3. Related work

We categorize the related work on fake news detection into three types: (1) Content-based Methods; (2) Knowledge-augmented Methods; (3) Explainable Methods.

3.1. Content-based fake news detection

Existing content-based methods currently in existence can be classified into two categories: unimodal methods (Castillo et al., 2011; Chen et al., 2018; Huh et al., 2018; Przybyla, 2020; Qi et al., 2019; Wu et al., 2015; Zhang et al., 2021) and multimodal methods (Jin et al., 2017; Khattar et al., 2019; Qi et al., 2021; Wu, Zhan, et al., 2021; Zhang et al., 2022). Unimodal methods perform fake news detection by employing techniques such as constructing feature words (Castillo et al., 2011), analyzing text sentiment (Wu et al., 2015), or mining writing styles (Przybyla, 2020). For instance, Castillo et al. (2011) utilize linguistic features including special characters, sentiment (positive/negative) words, emoticons, and others to identify fake news. Wu et al. (2015) propose a classifier based on a hybrid support vector machine to acquire textual sentiment semantic features for classifying false information. Multimodal methods conduct fake news detection by devising various sophisticated cross-modal fusion techniques. Wu, Zhan, et al. (2021) introduced a Co-Attention method (MCAN) to integrate textual and visual information for fake news detection. Li et al. (2021) propose a novel paradigm for detecting false information by aligning and fusing multimodal entities. Nevertheless, content-based methods overlook additional knowledge and exhibit moderate performance in the classification of meticulously designed fake news.

3.2. Knowledge-augmented fake news detection

In fake news detection task, additional knowledge primarily falls into two categories: social context (Dou et al., 2021, 2021; Ma, Gao, Wei, Lu, & Wong, 2015; Ren & Zhang, 2021; Sheng et al., 2022, 2022; Su et al., 2023; Yang, Liu, Yu, & Yang, 2012) and knowledge graphs (Ma, Gao, Joty, & Wong, 2019; Pan et al., 2018; Sun et al., 2023; Wang et al., 2020; Wu, Rao, Sun, & He, 2021; Wu, Rao, et al., 2021; Zhang, Zhang, Zhou, Huang, & Li, 2024). Methods based on social context utilize news propagation networks derived from users relevant to information on social media platforms, incorporating news-related posts and user-specific features for fake news detection. Ren and Zhang (2021) construct a heterogeneous information network based on information content, publishing users, and information topics, and utilized hierarchical attention mechanisms to learn node representations. Sheng et al. (2022) conduct fake news detection by mining features from all recent news and a subset of similar event news. Jin, Wang, Yang, Sun, Wang, Liao, and Xie (2022) employ inferential methods to unearth word-level fine-grained associative relationships among information-related posts for fake news detection. Methods grounded in knowledge graphs detect fake news by identifying conflicts between news content and validated knowledge. Pan et al. (2018) devised a technique that involves building an external knowledge graph derived from factual information pertaining to background topics found within the DBpedia dataset. Their approach centers around detecting fake news through the utilization of positive and negative knowledge graph embedding models. Hu et al. (2021) propose a novel model called CompareNet, which directly compares information with external knowledge for fake news detection. However, currently, no work integrates all available external knowledge into a unified framework, and existing models lack explainability.

3.3. Explainable fake news detection

To enhance the credibility of fake news detection outcomes and facilitate the understanding of model classification results, numerous studies are currently exploring methods for explainable fake news detection (Lourenço & Paes, 2022; Lu & Li, 2020; Shu et al., 2019; Yang et al., 2022). Current efforts in explainable fake news detection primarily focus on interpreting classification outcomes by extracting key information from single modalities. Shu et al. (2019) first address the explainability issue in fake news detection. They introduce a collaborative attention network between sentences and comments, aimed at capturing the top-k explainable aspects. Lu and Li (2020) employ attention mechanisms to identify suspicious nodes in the propagation network as part of the explanation. Yang et al. (2022) employ a distillation approach to select salient sentences from news content and comments. Lourenço and Paes (2022) directly consider retweet counts, comment data, and modality as factors for explanation. Thus, it is evident that the explanations generated by current methodologies are rather one-sided and fail to elucidate the distinct reasons behind the classification as true or false.

Diverging from prior research, our proposed MRE-FND model integrates multidimensional information, including news content, social networks, knowledge graphs, external news, etc. Utilizing information bottleneck theory to denoise various types of information, the model provides clear explanations for classification results by employing a key nodes set generation approach. Our approach addresses the inadequacy in leveraging news information and the poor explainability found in previous works, yielding superior results in experimental evaluations.

4. Problem formulation

A news is denoted as $n = \{T, I, U, C\}$, in which $T = \{t_1, t_2, \dots, t_n\}$ denotes the textual information, t_n denotes the sentence of text, I denotes the image information, $U = \{u_1, u_2, \dots, u_n\}$ denotes the users set which publish or forward news n , $C = \{c_1, c_2, \dots, c_n\}$ denotes the posts which related to n . The knowledge graph $\mathcal{KG} = \{(e_h, r, e_t) | e_h, e_t \in \mathcal{E}, r \in \mathcal{R}\}$ is structured as a collection of semantic triples, with $\mathcal{E}$ and $\mathcal{R}$ representing the sets of entities and relations, respectively. Our objective lies in learning a function f to effectively discriminate the fabricated news: $Y = f(n, \mathcal{KG})$, and output a nodes set S_n which to explain the reason for classification. Here, the binary output variable Y denotes the label assigned to the input news, with $Y = 0$ and $Y = 1$ signifying fake and real news, respectively. The symbols used in the paper are listed in Table 1.

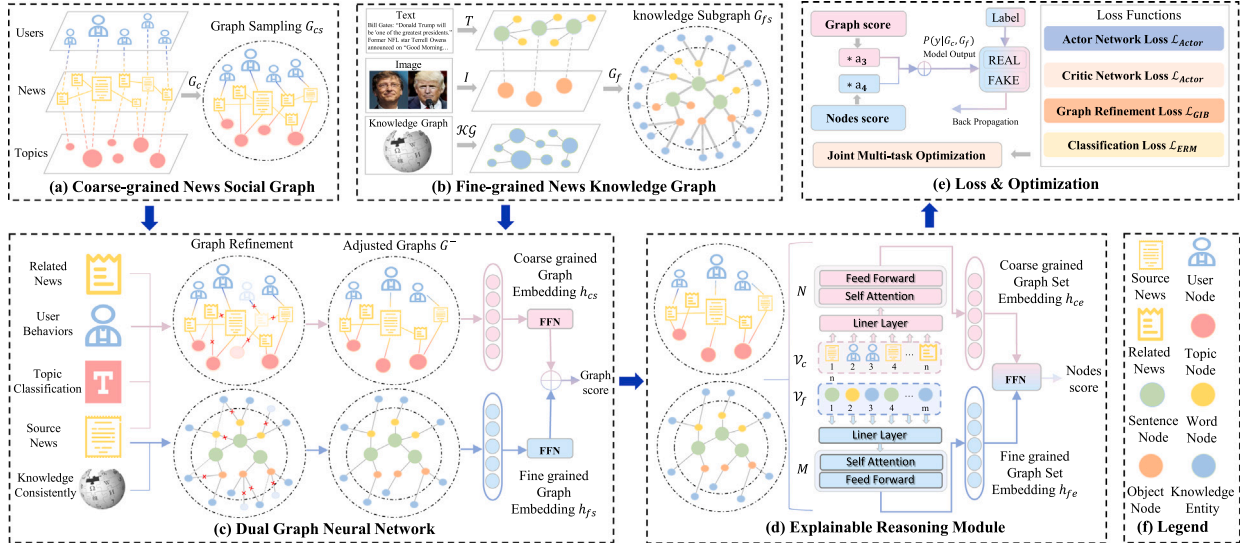


Fig. 2. The overview of Multifaceted Reasoning Network for Explainable Fake News Detection.

Table 1
Summary of notations.

Notation	Description
n	News containing text, images, users, and related posts
T	Textual information in the news
I	Image information in the news
U	Set of users related to the news
C	Set of posts related to the news
KG	Knowledge graph
$\mathcal{E}$	Set of entities in the knowledge graph
$\mathcal{R}$	Set of relations in the knowledge graph
G_c	Coarse-grained news social graph
G_{cs}	Coarse-grained news social subgraph
G_f	Fine-grained news knowledge graph
G_{fs}	Fine-grained news knowledge subgraph
s_t	State at step t
$\mathcal{A}_{s_t}$	Action space in state s_t
a_t	Action in Action space $\mathcal{A}_{s_t}$
G^-	Minimal graph of graph G
v	Node of Graph
h	Embedding of nodes
R_T	Terminal Reward
R_I	Immediate Reward
$\mathcal{T}$	Topic to which the news belongs
k	Keywords in the news
$\varphi(v_i)$	Set of l -order neighboring nodes of node v_i
S_{ens}	Explainable set of nodes
v_{sen}	Virtual node composed of sentence nodes

5. The proposed model

The overall architecture of the model we propose is shown in Fig. 2. Initially, we construct a coarse-grained social network graph for all news and a fine-grained news knowledge graph for individual news to respectively characterize the social features and content features, as shown in Fig. 2(a) and Fig. 2(b). The coarse-grained social network graph comprises user, source news, related posts, and news topic features. The fine-grained news content graph includes text, images, and knowledge graph features. Subsequently, we devised a dynamic graph neural network approach tailored for the dual-layer heterogeneous graph to update node features and adjacency matrices for reducing noise while preserving valuable information, as depicted in Fig. 2(c). Furthermore, we designed an inference module to generate explainable nodes sets on each graph for explaining the classification results, as depicted in Fig. 2(d). Finally, in order to train the model more efficiently, we design multiple losses and optimized the model by a joint multi-task optimization framework, as shown in Fig. 2(e). The details are described as follows.

5.1. Coarse-grained news social graph construction

In fake news detection, social relationships provide valuable classification information (Jin et al., 2022; Mehta, Pacheco, & Goldwasser, 2022; Su et al., 2023). Trustworthy users tend to share credible news, while unreliable users are more likely to spread misinformation. Related news can offer additional context. Prior research also highlights the importance of topic models in detecting fake news (Jin et al., 2022). Therefore, we created a heterogeneous graph at the news level. The graph, denoted as $G_c = (V_u, V_{ns}, V_{nr}, V_t, E_c)$, where V_u, V_{ns}, V_{nr}, V_t and E_c denote user nodes set, raw news set, related posts set, topic nodes set, and edges set, respectively. We included five types of edges: user-news publishing, user-news forwarding, related posts-source news, source news-topic, and related posts-topic.

We utilize CLIP (Radford et al., 2021b) to extract textual features from the raw news and related posts as the initial embeddings. The user's initial embedding is encoded using CLIP (Radford et al., 2021b) based on their historical posted posts. We employ an LDA (Blei, Ng, & Jordan, 2003) model to partition all texts into n_T topics. For each topic T , its representation is based on a distribution of keywords $P(k_i|T)$ within texts, where k_i denotes the i th keywords. The top N_K keywords for each topic T are determined as those with the highest values of $P(k_i|T)$. We encoded the topic words using BERT, and obtained the initial embeddings for each topic by probabilistically weighted summation.

The dissemination of news within social networks typically involves extensive user engagement and dense propagation of news articles. The coarse-grained news social graph we constructed may be considerably large in reality. To make large-scale graph learning feasible and accelerate the embedding learning process, we employ a graph sampling approach. Specifically, we center the sampling on the node of the news to be classified in the coarse-grained news social graph and collect its 3-order neighboring nodes to generate subgraphs G_{cs} . The graph sampling module addresses the neighborhood explosion problem caused by the virality of a news article or excessive user interactions in large-scale social networks.

5.2. Fine-grained news knowledge graph construction

The content of news articles is crucial for fake news classification, but relying solely on it yields limited results. As fake news becomes more sophisticated, external knowledge is often needed for detection. Thus, we take a fine-grained approach to build our model, considering sentences, entities in the text, objects in images, and nodes in a knowledge graph as graph nodes. The graph is denoted by $G_f = (V_s, V_w, V_o, V_{kg}, E_f)$, where V_s, V_w, V_o, V_{kg} , and E_f denote the sentences nodes set, text words nodes set, image objects nodes set, knowledge graph nodes set, and edges set, respectively. We constructed six types of edges: fully connected edges between sentences, co-occurrence relationships between sentences and text entities, sentences and image objects, text entities and knowledge graph nodes, image objects and knowledge graph nodes, and relationships in the knowledge graph.

For text, we use TagMe¹ to extract and link entities to a knowledge graph. For images, we use the Baidu AI Cloud API² to extract and link entities to the knowledge graph. An edge is constructed between a sentence and an image object based on the co-occurrence of textual entities and image entities. We used CLIP (Radford et al., 2021b) to extract features for each sentence as initial embeddings for sentence nodes and TransE for encoding other nodes linked to the knowledge graph.

Embedding a knowledge graph that encompasses the entire context of knowledge directly would introduce a significant amount of noise and irrelevant information. Furthermore, dealing with an extensive knowledge graph poses challenges for the iterative training of the model. Therefore, drawing inspiration from Jiang, Li, Cai, and Wang (2023) and Liu et al. (2021), we devise a reinforcement learning approach for knowledge graph sampling to select the most valuable knowledge. We model this process as a Markov Decision Process and utilize the A3C algorithm to absorb valuable nodes.

We define the state at step t of the Markov Decision Process as $s_t = (v_{sen}, \mathcal{E}_t, \mathcal{R}_t, E_t)$, where v_{sen} is the initial virtual node, $\mathcal{E}_t$ are the entities absorbed by previous actions, $\mathcal{R}_t$ are the relationships absorbed by previous actions, and E_t are the entities newly absorbed at step t . In state s_t , the agent selects an action from the action space $\mathcal{A}_{s_t} = \{(r_i, e_i) \mid (e_h, r_i, e_i) \in G_f, e_h \in E_t, e_i \notin \mathcal{E}_t\}$, incorporating the corresponding relationships and entities into the subgraph. Here (e_h, r_i, e_i) denotes the triplet of entity and relationships. We employ a policy network to compute the likelihood of the agent transitioning from one state to another after executing a specific action. The formulation is provided below:

$$P(s_{t+1} = (v_{sen}, \mathcal{E}_{t+1}, \mathcal{R}_{t+1}, E_{t+1}) \mid s_t, a_t) = 1 \quad (1)$$

where $s_{t+1} = (v_{sen}, \mathcal{E}_{t+1}, \mathcal{R}_{t+1}, E_{t+1})$ is the next state, and $a_t \in \mathcal{A}_{s_t}$. To provide feedback during agent-environment interaction, we devise two types of rewards: the Terminal Reward ($\mathcal{R}_T$), given at the final state, and the Immediate Reward ($\mathcal{R}_I$), provided at each state. Further details are in Section 5.5.1.

We implement the A3C (Mnih et al., 2016) algorithm to generate subgraphs, employing both an Actor and a Critic. We utilize the Actor network to calculate the probability of taking action a_t under state s_t , as shown in the following formula:

$$\pi_\beta(s_t, a_i) = \text{Softmax}\left(W_{\beta_1} \text{ReLU}\left(W_{\beta_2}[s_t; a_i]\right)\right) \quad (2)$$

where

$$s_t = [h_s; \mu_t; \bar{E}_t; \bar{R}_t], a_i = [r_i; e_i] \quad (3)$$

¹ <https://sobigdata.d4science.org/web/tagme/>

² <https://cloud.baidu.com/product/imagerecognition>

operator $[\cdot]$ denotes the concatenation operation, h_s denotes the sentence subgraph embedding, μ_t denotes the embedding of the knowledge subgraph at step t , $\bar{E}_t$, and $\bar{R}_t$ denote the average embeddings of the newly added entity and relationship respectively. The Critic network computes the Q-value to evaluate each action, as shown below:

$$Q_\theta(s_t, a_t) = \text{Sigmoid}\left(W_{\theta_1} \text{ReLU}\left(W_{\theta_2} [s_t; a_t]\right)\right) \quad (4)$$

At each step, we select the top- d actions with the highest probabilities, absorbing the corresponding entities and relationships into the subgraph. The value of d is top-5 for 1-hop, top-3 for 2-hop, and top-2 for 3-hop.

5.3. Dual graph neural network for fake news detection

We designed a dual graph neural network (DGNN) for both coarse-grained news social graph and fine-grained news knowledge graph to generate the graph representations. Manually defined graph construction strategies may result in the generation of redundant edges or overlook certain implicit edges. In coarse-grained news social graph, user behavior in social networks exhibits significant randomness, and not all users and news contribute to the classification of news. Moreover, in fine-grained news knowledge graph, not all entities and sentences contribute to news classification. In response to this issue, we employ a feature refinement method based on the graph information bottleneck principle (Sun et al., 2022; Wu, Fei, Cao, Bing, & Chua, 2023). In the following, we will provide a detailed introduction to our DGNN model. Firstly, we introduce the feature refinement process in Section 5.3.1. Then, the co-enhanced classification method is introduced in Section 5.3.2.

5.3.1. Feature refinement

We utilize feature refinement method to generate a minimal graph G^- , aiming to retain the original graph information while minimizing noise. And the same feature refinement method is employed for both the coarse-grained news social graph and the fine-grained news knowledge graph. Here, the approach is illustrated using the example of a coarse-grained news social graph.

Node Filtering. For each node v_i , we determine whether it is retained by sampling a value $\rho_i^v \in \{0, 1\} \sim \text{Bernoulli}(\pi_i^v)$ from a Bernoulli distribution. When $\rho_i^v = 0$, the node is discarded; when $\rho_i^v = 1$, it is retained. Here, $\pi_i^v \in (0, 1)$ is a parameter computed from the information of node v_i and its l -order neighborhood. We obtain the l -order context information for v_i through a weighted summation method. The complete process is shown as follows:

$$\begin{aligned} \alpha_{i,k}^v &= \frac{\exp\left(W_{\eta_1} [h_{v_i}; h_{v_k}]\right)}{\sum_{v_c \in \varphi(v_i)} \exp\left(W_{\eta_1} [h_{v_i}; h_{v_c}]\right)} \\ r_i^v &= \text{Tanh}\left(\sum_{v_k \in \varphi(v_i)} \alpha_{i,k}^v \left(W_{\eta_2} h_{v_k}\right)\right) \end{aligned} \quad (5)$$

where $\alpha_{i,k}^v$ denotes the weight of node v_k relative to node v_i , r_i^v aggregates the l -order neighborhood information of v_i , h_{v_i} denotes the embedding of v_i , $\varphi(v_i)$ denote the l -order neighbor nodes of v_i , W_{η_1} and W_{η_2} are parameter matrices. Then, we obtain π_i^v through a feedforward network:

$$\pi_i^v = \text{Sigmoid}\left(W_{\eta_3} \left(W_{\eta_4} [r_i^v; h_{v_i}] + b_{\eta_1}\right) + b_{\eta_2}\right) \quad (6)$$

where W_{η_3} , W_{η_4} , b_{η_1} , b_{η_2} are trainable parameters. However, the process of sampling ρ_i^v is discrete, we make it differentiable via the concrete relaxation method (Jang, Gu, & Poole, 2016).

$$\rho_i^v = \text{Sigmoid}\left(\frac{1}{\tau} \left(\log \frac{\pi_i^v}{1 - \pi_i^v} + \log \frac{\epsilon}{1 - \epsilon}\right)\right) \quad (7)$$

where τ is the temperature, $\epsilon \sim \text{Uniform}(0, 1)$.

Edge Adjusting. For each edge $e_{i,j}$, we employ the same approach to sample the value $\rho_{i,j}^e$ from a Bernoulli distribution. The parameter $\pi_{i,j}^e$ is computed from the node v_i, v_j , and their l -order neighborhoods. The complete process is shown as follows:

$$\pi_{i,j}^e = \text{Sigmoid}\left(W_{\eta_5} \left(W_{\eta_6} [r_i^e; h_{v_i}; r_j^e; h_{v_j}] + b_{\eta_3}\right) + b_{\eta_4}\right) \quad (8)$$

where r_i^e and r_j^e are obtained through Eq. (5), W_{η_5} , W_{η_6} , b_{η_3} , b_{η_4} are trainable parameters.

5.3.2. Co-enhanced classification

We perform feature refinement separately on the coarse-grained social news graph G_{cs} and the fine-grained news knowledge graph G_{fs} , resulting in adjusted graphs G_{cs}^- and G_{fs}^- , respectively. We utilize Graph Attention Network (GAT) to update the graph node representations and employ mean pooling to compute the graph representation vector. The formula for the coarse-grained social news graph G_{cs}^- is as follows:

$$\begin{aligned} \hat{X}_{cs}^- &= \text{GAT}\left(A_{cs}^-, X_{cs}^-\right) \\ h_{cs} &= W_{\eta_{cl}} \left(\text{Pooling}\left(\hat{X}_{cs}^-\right)\right) + b_{\eta_5} \end{aligned} \quad (9)$$

where A_{cs}^- and X_{cs}^- respectively denote the adjacency matrix and the feature matrix of the graph G_{cs}^- , $W_{\eta_{c1}}$ and b_{η_5} are trainable parameters. The computation process for the fine-grained news knowledge graph is similar to the Eq. (9).

$$\begin{aligned}\hat{X}_{fs}^- &= \text{GAT}(A_{fs}^-, X_{fs}^-) \\ h_{fs} &= W_{\eta_{f1}}(\text{Pooling}(\hat{X}_{fs}^-)) + b_{\eta_6}\end{aligned}\quad (10)$$

The final classification prediction results for DGNN are expressed by the following formula:

$$\begin{aligned}p_{d_{cs}} &= \text{Sigmoid}(W_{\eta_{c2}}h_{cs} + b_{\eta_7}) \\ p_{d_{fs}} &= \text{Sigmoid}(W_{\eta_{f2}}h_{fs} + b_{\eta_8}) \\ p_d &= \alpha_1 \times p_{d_{cs}} + (1 - \alpha_1) \times p_{d_{fs}}\end{aligned}\quad (11)$$

where α_1 is hyperparameter, $W_{\eta_{c2}}, W_{\eta_{f2}}, b_{\eta_7}, b_{\eta_8}$ are trainable parameters.

5.4. Explainable reasoning module

Existing fake news detection methods lack explanations for their classification results. Moreover, the explanations generated by a few explainable fake news detection methods are often quite limited. To address this issue, we have designed an explainable reasoning module (ERM) that provides explanations for the news classification results, while simultaneously improving the accuracy of news classification. Specifically, the ERM is divided into two parts. Firstly, we introduce the method for generating key node sets in Section 5.4.1, and then, in Section 5.4.2, we present the classification method based on the key node sets.

5.4.1. Key nodes sets generation

Based on the optimized graph structure, we have devised a method for generating key node sets using Eigenvector Centrality, extracting key nodes for fake news classification. We consider the connectivity of central nodes and the connectivity of their neighboring nodes to determine their importance scores. The calculation formula is as follows:

$$\text{Score}(v_i) = D(v_i) + \sum_{v_j \in \mathcal{V}(v_i)} D(v_j) \quad (12)$$

where $D(v_i)$ refers to the connectivity of node v_i , and $\mathcal{V}(v_i)$ refers to the neighborhood nodes of v_i . For different categories of nodes, we respectively select the top- k nodes with the highest scores as key nodes for subsequent classification.

5.4.2. Classification method based on key nodes sets

We utilize the obtained key nodes sets to classify fake news. Due to the interconnections among different nodes and the varying importance of relationships between them, we employ the multi-head attention (Vaswani et al., 2017) to aggregate node features. The formula for basic self-attention is presented as follows:

$$\hat{H}^{(l)} = \text{SelfAtt}(Q^{(l)}, K^{(l)}, V^{(l)}) = \text{softmax}\left(\frac{Q^{(l)}K^{(l)\top}}{\sqrt{d_k}}\right)V^{(l)} \quad (13)$$

where

$$\begin{cases} Q^{(l)} = H^{(l-1)}W_{Q_1}^{(l)}, \\ K^{(l)} = H^{(l-1)}W_{K_1}^{(l)}, \\ V^{(l)} = H^{(l-1)}W_{V_1}^{(l)}. \end{cases} \quad (14)$$

Matrices $W_{Q_1}^{(l)}, W_{K_1}^{(l)}, W_{V_1}^{(l)} \in \mathbb{R}^{d \times d_k}$ are trainable parameters. Multi-head attention divides $Q^{(l)}, K^{(l)}$, and $V^{(l)}$ into multiple heads, each performing self-attention processing independently, and subsequently combines the outputs linearly to obtain the result. The feed-forward layer is calculated as follow:

$$H^{(l)} = \text{FeedForward}(\hat{H}^{(l)}) = \text{ReLU}(\hat{H}^{(l)}W_{\eta_1}^{(l)} + b_{\eta_1}^{(l)})W_{\eta_2}^{(l)} + b_{\eta_2}^{(l)} \quad (15)$$

where $W_{\eta_1}^{(l)}, W_{\eta_2}^{(l)}, b_{\eta_1}^{(l)}, b_{\eta_2}^{(l)}$ are trainable parameters.

For the coarse-grained graph nodes set $\mathcal{V}_c$, the forward process is formalized as:

$$\begin{aligned}\hat{H}_c^{(l)} &= \text{LN}(\text{SelfAtt}(Q_c^{(l)}, K_c^{(l)}, V_c^{(l)}) + H_c^{(l-1)}) \\ H_c^{(l)} &= \text{LN}(\text{FeedForward}(\hat{H}_c^{(l)}) + \hat{H}_c^{(l)}).\end{aligned}\quad (16)$$

where

$$H_c^{(0)} = \text{Concat}(h_{v_{c1}}, h_{v_{c2}}, \dots, h_{v_{c_n}})_{v_{c_n} \in \mathcal{V}_c} \quad (17)$$

Vectors $h_{v_{c1}}, h_{v_{c2}}, \dots, h_{v_{cn}} \in \mathbb{R}^d$ denotes the embeddings of coarse-grained key nodes. The same process is applied to the fine-grained key nodes set $\mathcal{V}_f$, which is formulated as follows:

$$\begin{aligned}\hat{H}_f^{(l)} &= \text{LN}(\text{SelfAtt}(Q_f^{(l)}, K_f^{(l)}, V_f^{(l)}) + H_f^{(l-1)}) \\ H_f^{(l)} &= \text{LN}(\text{FeedForward}(\hat{H}_f^{(l)}) + \hat{H}_f^{(l)}).\end{aligned}\quad (18)$$

where

$$H_f^{(0)} = \text{Concat}(h_{v_{f1}}, h_{v_{f2}}, \dots, h_{v_{fn}})_{v_{fn} \in \mathcal{V}_f} \quad (19)$$

Vectors $h_{v_{f1}}, h_{v_{f2}}, \dots, h_{v_{fn}} \in \mathbb{R}^d$ denotes the embeddings of fine-grained key nodes. The embedding of [CLS] token from the top layer is the final representation of coarse-grained key nodes set and fine-grained key nodes set. The final classification prediction results for ERM are expressed by the following formula:

$$p_e = \text{Sigmoid}\left(W_{\eta_3} [h_{ce}; h_{fe}] + b_{\eta_3}\right) \quad (20)$$

where h_{ce} and h_{fe} denote the final representation of coarse-grained key nodes set and fine-grained key nodes set, W_{η_3} and b_{η_3} are trainable parameters. The model ultimately considers the predictions of both DGNN and ERM as shown in the following equation:

$$p = \alpha_2 \times p_d + (1 - \alpha_2) \times p_e \quad (21)$$

where α_2 is hyperparameter, p_d and p_e denote the classification prediction results of DGNN and ERM, respectively.

5.5. Loss & optimization

For efficient model training, we utilize a joint multi-task optimization framework to optimize the model. In this section, we first introduce the reinforcement learning reward function used for generating knowledge subgraphs, then present various losses employed for model optimization, and finally discuss the joint multi-task optimization framework designed by us.

5.5.1. Implementation of rewards

As described in Section 5.2, we design two reward functions: Terminal Reward and Immediate Reward. In this section, we provide a detailed exposition of these two types of rewards.

Terminal Reward is utilized to assess the suitability of the generated subgraph for the downstream task. In the DGNN module, we utilize the generated news subgraphs to represent news. Therefore, we employ a function that is opposite to cross-entropy to measure whether the generated subgraph contributes to fake news detection. The calculation formula is as follows:

$$\mathcal{R}_T^D = Y \cdot \log(p_d) + (1 - Y) \cdot \log(1 - p_d) \quad (22)$$

where Y is the classification label of the news, p_d is derived from Eq. (11). In the ERM module, we utilize the generated news subgraphs to seek explainable node sets for fake news detection. Therefore, for the ERM module, we also employ a function opposite to cross-entropy to measure whether the generated subgraph contributes to fake news detection. The calculation formula is as follows:

$$\mathcal{R}_T^E = Y \cdot \log(p_e) + (1 - Y) \cdot \log(1 - p_e) \quad (23)$$

where p_e is derived from Eq. (20). The final terminal reward is the weighted sum of two parts of rewards. The calculation formula is as follows:

$$\mathcal{R}_T = \alpha_3 \times \mathcal{R}_T^D + (1 - \alpha_3) \times \mathcal{R}_T^E \quad (24)$$

Immediate Reward: Terminal Reward can only be calculated after all actions have been completed and cannot provide feedback to the agent after each action. To address this issue, we employ an Immediate Reward. We encourage the agent to select entities which have similar semantic topics (we call it coherence) to the current news article n_a . To this end, we build a reverse index to track all the articles that mention an entity e in the news: $\Gamma(e) = \{n_1, n_2, \dots, n_k \mid n_i \text{ contains } e\}$. The coherence reward:

$$\mathcal{R}_I(n_a, e) = \text{cosine}\left(h_{n_a}, \frac{\sum_{n_i \in \Gamma(e)} h_{n_i}}{|\Gamma(e)|}\right) \quad (25)$$

where h_{n_a} and h_{n_i} denote the embeddings of news n_a and news n_i , respectively. The Immediate Reward for each action is aggregated to obtain the final Immediate Reward, as depicted by the following formula:

$$\mathcal{R}_I = \sum_{e_i \in E_m} (\mathcal{R}_I(n_a, e_i)) \quad (26)$$

where E_m denotes the selected set of entities. Finally, we aggregate the Terminal Reward and Immediate Reward by weighted summation to obtain the final reward function.

$$\mathcal{R}(t) = \alpha_4 \times \mathcal{R}_T(t) + (1 - \alpha_4) \times \mathcal{R}_I(t) \quad (27)$$

5.5.2. Loss function

In this section, we introduced the loss functions used for model optimization. We designed specific loss functions for the reinforcement learning module, dual graph neural network module, and explainable reasoning module, respectively. Detailed explanations as follow.

The critic network undergoes training utilizing the TD error expectation (Sutton, 1988), and the actor network is trained with policy gradient (Sutton, McAllester, Singh, & Mansour, 1999). The formula is presented as follows:

$$\begin{aligned}\mathcal{L}_{\text{Critic}}(\theta) &= \mathbb{E}_{a \sim \pi_{\theta}} \left[(Q_{\theta}(s_t, a) - q_t)^2 \right], \\ \mathcal{L}_{\text{Actor}}(\beta) &= \mathbb{E}_{a \sim \pi_{\beta}} \left[(Q_{\theta}(s_t, a) - q_t) * \pi_{\beta}(s_t, a) \right],\end{aligned}\quad (28)$$

where θ and β represent the parameters, $Q_{\theta}(s_t, a)$ and $\pi_{\beta}(s_t, a)$ are as shown in Eqs. (2) and (4), respectively. q_t denotes the target value calculated using the Bellman Eq. (29)

$$q_t = \mathcal{R}(t) + \mathbb{E}_{a \sim \pi_{\beta}} [\gamma \cdot Q_{\theta}(s_{t+1}, a)]. \quad (29)$$

In order to optimize graphs G_{cs} and G_{fs} to remove as much noise as possible while retaining useful information for fake news classification, we employ a GIB-guided optimization method. The optimization objective is as follows:

$$\begin{aligned}\mathcal{L}_{\text{GIB}}^{cs} &= \min_{z_{cs}} [-I(z_{cs}, Y) + \beta \cdot I(z_{cs}, G_{cs})], \\ \mathcal{L}_{\text{GIB}}^{fs} &= \min_{z_{fs}} [-I(z_{fs}, Y) + \beta \cdot I(z_{fs}, G_{fs})].\end{aligned}\quad (30)$$

where z_{cs} and z_{fs} are sampled from a Gaussian distribution parameterized by a_{cs} and a_{fs} , and $-I(z, Y)$ is calculated as follow:

$$\begin{aligned}-I(z, Y) &\leq -\int p(Y, z) \log q(Y | z) dY dz + H(Y) \\ &:= \mathcal{L}_{\text{CE}}(q(Y | z), Y),\end{aligned}\quad (31)$$

where $q(Y | z)$ is a variational approximation of the true posterior $p(Y, z)$, $H(Y)$ denotes the Shannon entropy of Y , and we estimate the upper bound of $I(z, G)$ by reparameterization trick (Kingma & Welling, 2014):

$$\begin{aligned}I(z, G) &\leq \int p(z | G) \log \frac{p(z | G)}{r(z)} dz dG \\ &:= \text{KL}(p(z | G) \| r(z)).\end{aligned}\quad (32)$$

where $r(z)$ is the variational approximation to the prior distribution $p(z)$ of z .

Finally, to optimize the classification performance of the explainable reasoning module, we employ a cross-entropy loss function for optimization. The specific formula is as follows:

$$\mathcal{L}_{\text{ERM}} = -(Y \cdot \log(p_e) + (1 - Y) \cdot \log(1 - p_e)) \quad (33)$$

where p_e is derived from Eq. (20).

5.5.3. The optimization framework

Our model consists of three main components with trainable parameters, which are both independent and interdependent. The purpose of joint optimization is to optimize each module while allowing downstream modules to guide upstream modules optimization to support downstream fake news detection and explainable node set generation tasks. Our optimization framework focuses on the relationships between different modules. Our algorithm is presented in Algorithm 1. We first construct graph, as shown in lines 3–5. Next, the Dual Graph Neural Network is employed to optimize the graph structure and classify news articles at two different granularities, as shown in lines 6–8. Then, the Explainable Reasoning Module is utilized for generating explanations, as shown in lines 9–13. Finally, the Joint Multi-task Optimization framework is employed for efficiently training the model, as shown in lines 14–16.

6. Experiments

To ascertain the effectiveness of the proposed model, we conduct some experiments. This section provides a detailed introduction to the experiments.

6.1. Experimental settings

Datasets. We conduct experiments on three commonly employed datasets: Politifact (Shu, Mahudeswaran, Wang, Lee, & Liu, 2020), Gossipcop (Shu et al., 2020) and PHEME (Zubiaga, Liakata, & Procter, 2017). PolitiFact and GossipCop originate from fact-checking platforms established to verify the credibility of disseminated news. These datasets include user information, related posts, as well as images and text of the news. To evaluate the model's performance on datasets with limited information, we incorporated the PHEME dataset collected from Twitter, which contains only images and text, without any social network information. we remove the missing and low-quality image data and perform data cleaning on all the text data to eliminate redundant and irrelevant information.

Algorithm 1: The calculation process of MRE-FND

Input : Raw News set S_n , User Set S_u , Related Post set S_{rp} , Knowledge Graph $\mathcal{KG}$;
output: Classification prediction p , Explainable nodes Set S_{ens} , Parameters of subgraph generation network (θ and β), Parameters of dual graph neural network (η_c and η_f), Parameters of explainable reasoning module (δ);

```

1 Select  $B$  samples;
2 for  $b \leftarrow 1$  to  $B$  do
3   /* Construct Graph                                     */
4    $G_{cs} \leftarrow \text{CoarseConstructGraph}(S_n, S_u, S_{rp})$ ;
5    $G_{fs} \leftarrow \text{FineConstructGraph}(N_b, \mathcal{KG})$ ;
6    $\mathcal{R}_l \leftarrow \text{Calculate Immediate Reward}$ ;
7   /* Dual Graph Neural Network for Fake News Detection */
8    $G_{cs}^- \leftarrow \text{CoarseGraphRefine}(G_{cs})$ ;
9    $G_{fs}^- \leftarrow \text{FineGraphRefine}(G_{fs})$ ;
10   $p_{cs}, p_{fs}, p_d \leftarrow \text{CoEnhanceClassification}(G_{cs}^-, G_{fs}^-)$ ;
11  /* Explainable Reasoning Module                         */
12   $\mathcal{V}_c, \mathcal{V}_f \leftarrow \text{KeyNodesSetsGeneration}(G_{cs}^-, G_{fs}^-)$ ;
13   $H_c \leftarrow \text{SelfAttforCoarse}(\mathcal{V}_c)$ ;
14   $H_f \leftarrow \text{SelfAttforFine}(\mathcal{V}_f)$ ;
15   $p_e \leftarrow \text{NodesSetClassification}(H_c, H_f)$ ;
16   $\mathcal{R}_T^p, \mathcal{R}_T^E \leftarrow \text{Calculate Terminal Rewards}$ ;
17  /* Joint Multi-task Optimization                       */
18  Freeze  $\theta, \beta, \eta_c$ , and  $\eta_f$  optimize parameters  $\delta$  with Eq. (33);
19  Freeze  $\theta, \beta$ , and  $\delta$  optimize parameters  $\eta_c$  and  $\eta_f$  with Eq. (30);
20  Freeze  $\eta_c, \eta_f$ , and  $\delta$  optimize parameters  $\theta$  and  $\beta$  with Eq. (27);
21 end
22 return  $p, S_{ens}, \theta, \beta, \eta_c, \eta_f, \delta$ 

```

Table 2
The statistics of the two datasets.

Dataset	User	Source news	Real news	Fake news	Related news	Words per news
PolitiFact	151 173	495	286	209	339 720	1546.1
GossipCop	303 844	15 707	12 154	3553	2 325 812	578.6
Pheme	–	2189	1549	640	–	13.8

Table 2 presents the basic information of the two datasets. Finally, the dataset is randomly split into training, validation, and testing sets with a ratio of 7:1:2. The knowledge graph in our experiment is constructed from the subset of Wikidata. Firstly, all entities in the news are extracted. Subsequently, we expand central entities by including connecting entities within 1-hop and 2-hop distances from Wikidata. Then, we gather the relations linking these entities and merge them into triples. After that, we will get a task-relevant knowledge graph from Wikidata. The knowledge graph contains 324173 entities, 783404 triples, and 1059 relations.

Baselines. Given that we address the problem without employing large language models, there is no comparison with the latest large language model baselines like GPT. We compare our model in two categories of state-of-the-art baselines, a total of eleven models. And the modalities used by different models are summarized in **Table 3**.

The unimodal fake news detection models:

- **VGG19** (Simonyan & Zisserman, 2014) is a prevalent CNN utilized for image classification.
- **TextCNN** (Chen, 2015) uses convolutional neural networks to extract text features and classify.
- **BERT** (Devlin, Chang, Lee, & Toutanova, 2018) is a pretrained model to extract text features.

The multimodal fake news detection models:

- **EANN** (Wang et al., 2018) designs a multi-task learning framework to detect fake news and classify events simultaneously.
- **MVAE** (Khatter et al., 2019) comprises an encoder, a decoder and a detector to classify fake news.
- **COOLANT** (Wang et al., 2023) employs contrastive learning to align images and text for fake news detection.
- **FSRU** (Lao et al., 2024) attempts to perform multimodal rumor detection in the frequency domain, transforming spatial features into spectral features for multimodal representation and fusion.
- **ALGM** (Dong et al., 2023) proposes a graph-theoretic framework called the Generalized Deep Markov Random Field framework, which inherits the capabilities of deep learning while leveraging the correlations among news articles.

Knowledge-enhanced fake news detection models:

- **dFEND** (Shu et al., 2019) classifies news articles by mining the most important sentences and comments within them.
- **CompareNet** (Hu et al., 2021) constructs a directed heterogeneous document graph to utilize knowledge base.

Table 3

The modalities used by different models.

Method	Unimodal Method			Multimodal Method					Knowledge Enhanced			Ours
	VGG-19	TextCNN	BERT	EANN	MVAE	COOLANT	FSRU	ALGM	dEFEND	CompNet	KAHAN	
Image	✓			✓	✓	✓	✓	✓				✓
Text		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
KG										✓	✓	✓
User											✓	✓
RN									✓			✓
Topic										✓		✓

Table 4Overall performance of different detection models. Best results are in **bold** and second best results are underlined.

Category	Method	Politifact				Gossipcop				PHEME			
		Acc	Pre	Rec	F1	Acc	Pre	Rec	F1	Acc	Pre	Rec	F1
Unimodal Method	VGG-19	0.515	0.582	0.561	0.571	0.543	0.735	0.637	0.682	0.523	0.688	0.597	0.639
	TextCNN	0.677	0.719	0.719	0.719	0.693	0.802	0.798	0.800	0.568	0.769	0.558	0.647
	BERT	0.808	0.806	0.877	0.840	0.814	0.869	0.893	0.881	0.753	0.861	0.777	0.817
Multimodal Method	EANN	0.768	0.804	0.789	0.796	0.734	0.810	0.856	0.832	0.767	0.859	0.803	0.830
	MVAE	0.818	0.855	0.825	0.839	0.785	0.852	0.872	0.862	0.758	0.821	0.842	0.831
	COOLANT	0.879	0.895	0.895	0.895	<u>0.826</u>	<u>0.871</u>	0.909	<u>0.890</u>	0.833	0.899	<u>0.861</u>	0.880
	FSRU	<u>0.909</u>	<u>0.914</u>	<u>0.930</u>	<u>0.922</u>	0.820	0.863	0.911	0.886	<u>0.836</u>	<u>0.905</u>	0.858	<u>0.881</u>
	ALGM	0.859	0.864	0.895	0.879	0.805	0.859	0.893	0.876	0.815	0.886	0.848	0.867
Knowledge Enhanced	dEFEND	0.869	0.893	0.877	0.885	0.823	0.862	<u>0.917</u>	0.888	–	–	–	–
	CompNet	0.818	0.842	0.842	0.842	0.791	0.865	<u>0.864</u>	0.864	0.801	0.881	0.832	0.856
	KAHAN	0.889	0.911	0.895	0.903	0.821	0.864	0.911	0.887	–	–	–	–
Proposed	MRE-FND	0.929	0.931	0.947	0.939	0.834	0.873	0.918	0.894	0.847	0.909	0.871	0.89
	Imp.(%)	+2.0	+1.7	+1.7	+1.7	+0.8	+0.2	+0.1	+0.4	+1.1	+0.4	+1.0	+0.9

- **KAHAN** (Tseng, Yang, Wang, & Peng, 2022) incorporates user comments, external knowledge, and temporal information to classify news.

Implementation Details. Our experiments utilize an RTX 3090 graphics card equipped with 24GB of memory. We leverage CLIP (Radford et al., 2021a) as the pre-trained model for embedding both text and images. The optimization is performed using the Adam algorithm (Kingma & Ba, 2014), with a batch size of 64. The initial learning rate is established at $2e^{-4}$. We set the embedding size for entities, relations, and documents at 1024. For the classification component within our explainable reasoning framework, we use a 2-layer standard Transformer encoder, configured with four attention heads.

6.2. Quantitative evaluation

We evaluate fake news detection results with the metrics of Accuracy, Precision, Recall and F1-score. We replicated all baseline models and conducted experiments on our dataset. The experimental result is detailed in Table 4. Our model achieved state-of-the-art performance across all metrics on the three datasets. On the Politifact dataset, the four metrics improve by 2.0, 1.7, 1.7, and 1.7 percentage points, respectively. On the Gossipcop dataset, the four metrics improve by 0.8, 0.2, 0.1, and 0.4 percentage points, respectively. On the PHEME dataset, the four metrics improved by 1.1, 0.4, 1.0, and 0.9 percentage points, respectively. Additionally, we observe that the text-based methods (TextCNN and BERT) outperform the visual-based ones (VGG-19), indicating the greater significance of textual semantics. Both multimodal approaches and knowledge-enhanced methods yield superior results compared to single-modal fake news detection methods. This corroborates our hypothesis that leveraging information from diverse sources enhances the performance of fake news detection. However, both multimodal approaches and knowledge-enhanced methods yielded results inferior to our MRE-FND model, demonstrating the effectiveness of our model in integrating fine-grained and coarse-grained features within news articles. The excellent results on the PHEME dataset demonstrate that our model performs well on datasets lacking social network information, indicating good generalization ability. In summary, MRE-FND effectively integrates various information sources including text, images, knowledge graphs, user data, related posts, and news topics, ultimately achieving outstanding comprehensive performance.

6.3. Ablation study

In this section, we conduct ablation studies to evaluate the significance of various modules. Table 5 presents the results from these different ablation models.

- **Modality Input.** Our objective is to assess the impact of various modalities. The labels “w/o Image,” “w/o KG,” “w/o User,” “w/o RP,” and “w/o Topic” denote inputs devoid of images, knowledge graphs, user data, related posts, and topics, respectively.

Table 5
Ablation studies of MRE-FND on two datasets.

Category	Method	Content Type						Politifact				Gossipcop			
		Image	Text	KG	User	RP	Topic	Acc	Pre	Rec	F1	Acc	Pre	Rec	F1
Proposed	MRE-FND	✓	✓	✓	✓	✓	✓	0.929	0.931	0.947	0.939	0.834	0.873	0.918	0.894
Modality Input	w/o Image		✓	✓	✓	✓	✓	0.919	0.930	0.930	0.930	0.831	0.873	0.914	0.893
	w/o KG	✓	✓		✓	✓	✓	0.899	0.912	0.912	0.912	0.807	0.871	0.914	0.893
	w/o User	✓	✓	✓		✓	✓	0.909	0.929	0.912	0.920	0.816	0.867	0.899	0.883
	w/o RP	✓	✓	✓	✓		✓	0.919	0.945	0.912	0.929	0.821	0.868	0.905	0.886
	w/o Topic	✓	✓	✓	✓	✓		0.919	0.930	0.930	0.930	0.824	0.873	0.903	0.888
Graph Ablation	w/o CGG	✓	✓	✓				0.889	0.944	0.895	0.919	0.807	0.869	0.883	0.876
	w/o FGG		✓		✓	✓	✓	0.889	0.926	0.877	0.901	0.809	0.869	0.886	0.877
	w/o RSG	✓	✓	✓	✓	✓	✓	0.899	0.898	0.930	0.914	0.812	0.867	0.893	0.880
Module Ablation	w/o DGNN	✓	✓	✓	✓	✓	✓	0.869	0.893	0.877	0.885	0.802	0.869	0.875	0.872
	w/o ERM	✓	✓	✓	✓	✓	✓	0.889	0.911	0.895	0.903	0.819	0.877	0.890	0.883

The experimental findings reveal that the models labeled “w/o User” and “w/o KG” demonstrate the most significant decrease in performance, underscoring the vital roles of user data and knowledge graphs in the task of fake news detection. Our analysis supports this observation, suggesting that credible users are more likely to share authentic news, whereas non-credible users are inclined to spread misinformation. The knowledge graph contributes nuanced information critical for discerning fake news. Additionally, the performance declines observed in the models “w/o Image”, “w/o KG”, “w/o User”, “w/o RP”, and “w/o Topic” compared to the MRE-FND model further affirm the value of each modality in enhancing detection capabilities.

• **Graph Ablation.** We conduct ablation experiments to evaluate the coarse-grained social news graph, fine-grained news knowledge graph, and knowledge-enhanced subgraph, assessing their individual contributions to our system’s performance. “w/o CGG” refers to the absence of the coarse-grained social news graph, “w/o FGG” indicates the absence of the fine-grained news knowledge graph, and “w/o RSG” denotes the exclusion of reinforcement learning for knowledge selection, opting instead to directly select the first-order neighborhood from the knowledge graph as external knowledge. Based on the experimental results, it can be observed that the contributions of the coarse-grained graph and the fine-grained graph to the classification outcomes exhibit similarities. Furthermore, the performance decline of “w/o RSG” relative to MRE-FND demonstrates the effectiveness of utilizing reinforcement learning to generate knowledge subgraph, which can effectively reduce noise.

• **Module Ablation.** Finally, we conduct ablation experiments on the two main modules of the model, the Dual Graph Neural Network and the Explainable Reasoning Module. “w/o DGNN” refers to the absence of the Dual Graph Neural Network module, while “w/o ERM” indicates the absence of the Explainable Reasoning Module. From the experimental results, it can be observed that “w/o GLM” exhibits a decrease of 6.0% compared to MRE-FND, demonstrating that the Dual Graph Neural Network effectively optimizes the graph structure and performs fake news classification. “W/o ERM” exhibits a decrease of 4.0% compared to MRE-FND, demonstrating that the Explainable Reasoning Module effectively extracts key features.

6.4. Parameter sensitivity analysis

In this section, we assess the influence of hyperparameters on the performance of the model. These hyperparameters include parameter α_1 (Eq. (11)) in the Dual Graph Neural Network module, which balances coarse-grained social network graph information and fine-grained news content graph information, parameter α_2 (Eq. (21)) in the final classification module, which balances the Dual Graph Neural Network module and Explainable Reasoning module, parameter α_3 (Eq. (24)) in the terminal reward, which balances the Dual Graph Neural Network module and Explainable Reasoning module, parameter α_4 (Eq. (27)) in the reward function, which balances Immediate Reward and Terminal Reward, and parameter n in Section 5.1, which used to generate subgraphs. Firstly, we systematically varied the four parameters $\alpha_1, \alpha_2, \alpha_3$, and α_4 from 0 to 1, with a step size of 0.1. Fig. 3(a–d) illustrates the curves depicting the variations in model metrics as a function of these hyperparameters. With the increase in balancing parameters, the model accuracy generally exhibits a trend of initial ascent followed by a decline. Specifically, on the Politifact dataset, the model achieves optimal results when $\alpha_1 = 0.5, \alpha_2 = 0.7, \alpha_3 = 0.7$ and $\alpha_4 = 0.8$. On the Gossipcop dataset, the optimal results are obtained when $\alpha_1 = 0.5, \alpha_2 = 0.7, \alpha_3 = 0.6$ and $\alpha_4 = 0.8$. Then, we varied the parameter n from 1 to 5, with a step size of 1. Fig. 3(e) displays the curves of accuracy as a function of n on the Politifact and Gossipcop datasets. As n increases, the accuracy first rises and then declines, achieving optimal results at $n = 3$. This aligns with our understanding that larger subgraphs can capture more information, but they also introduce significant noise. Thus, $n = 3$ is a reasonable value. Moreover, overall, the model accuracy does not exhibit significant fluctuations with the variations in hyperparameters, demonstrating the robustness of our model.

6.5. Evaluation and validation across datasets

To further verify the generalization ability of our model, we designed a cross-dataset experiment. Specifically, we randomly select 173 samples from both the Politifact and Gossipcop datasets to train our model. Subsequently, we test it on a Politifact test set comprising 99 samples. The results are compared with those of a model trained on a training set consisting of 346 samples

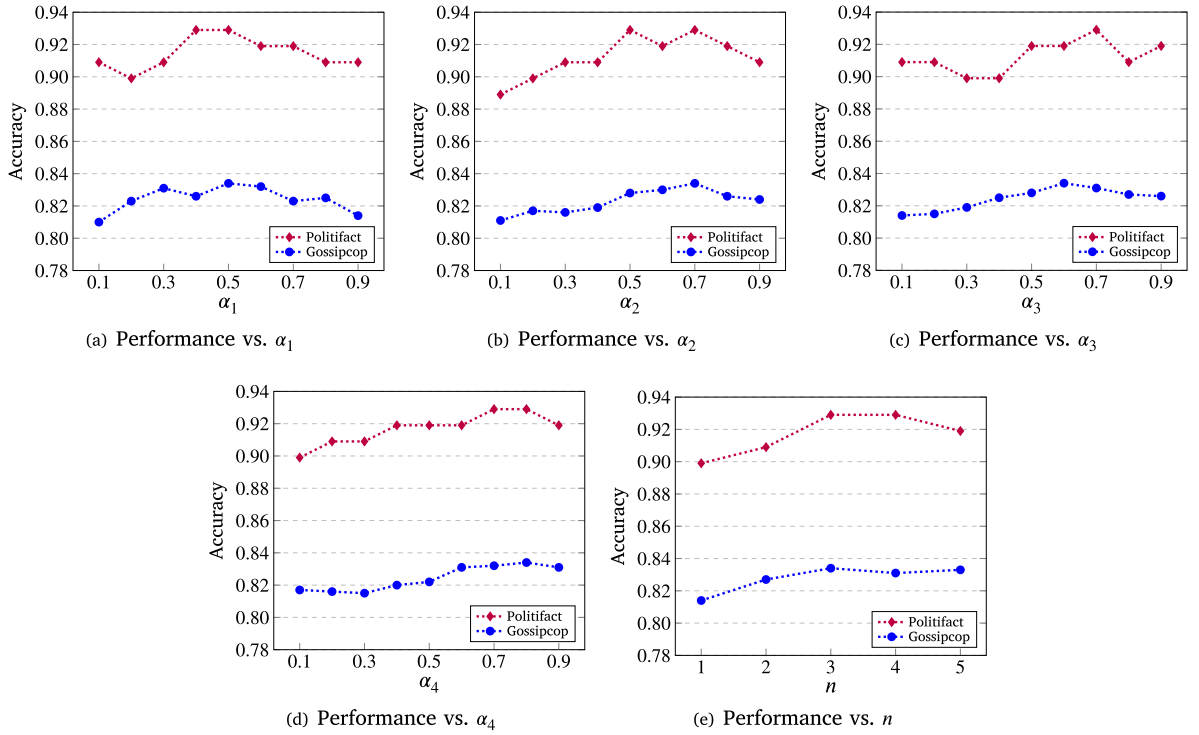


Fig. 3. The parameter sensitivity analysis experimental results for MRE-FND.

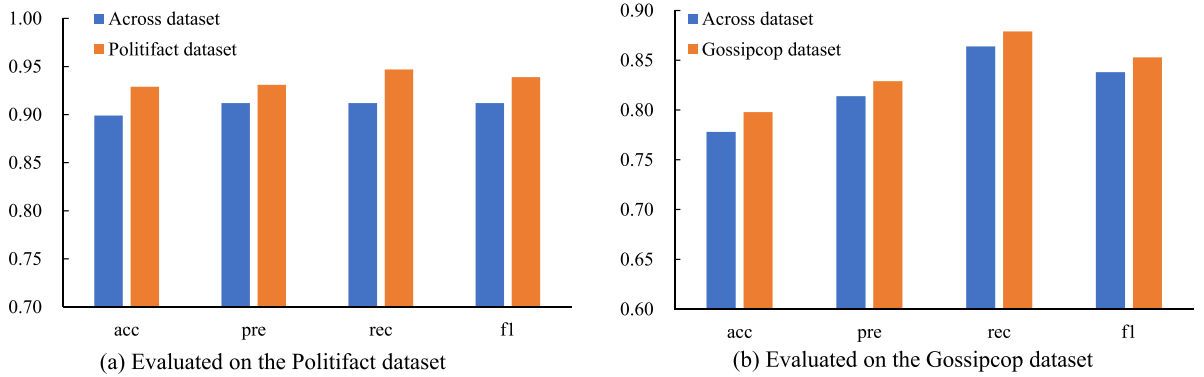


Fig. 4. Evaluation and validation across datasets.

exclusively from the Politifact dataset. As shown in Fig. 4(a), it can be observed that the performance of the model trained across datasets is comparable to that of the model trained solely on the Politifact dataset, demonstrating the generalization capability of our model. Similarly, we randomly select 173 samples from both the Politifact and Gossipcop datasets to train our model. We then test this model on a Gossipcop test set that included 99 samples, comparing the results with those from a model trained on a training set of 346 samples exclusively from the Gossipcop dataset. As illustrated in Fig. 4(b), similar to the results in Fig. 4(a), there is little difference between the two outcomes. This further demonstrates the generalization capability of our model.

6.6. Explainability analysis

In Fig. 5, we illustrate two test instances to demonstrate the coarse-grained and fine-grained key nodes generated by our method. In the first example, we accurately identified key entities 'Miley Cyrus' and 'Liam Hemsworth' within the fine-grained news knowledge graph. The knowledge graph contains the triples (Miley Cyrus, spouse, Liam Hemsworth) and (Liam Hemsworth, spouse, Miley Cyrus), which effectively explain the veracity of the news article 'Miley Cyrus Confirms She's Engaged to Liam Hemsworth'.

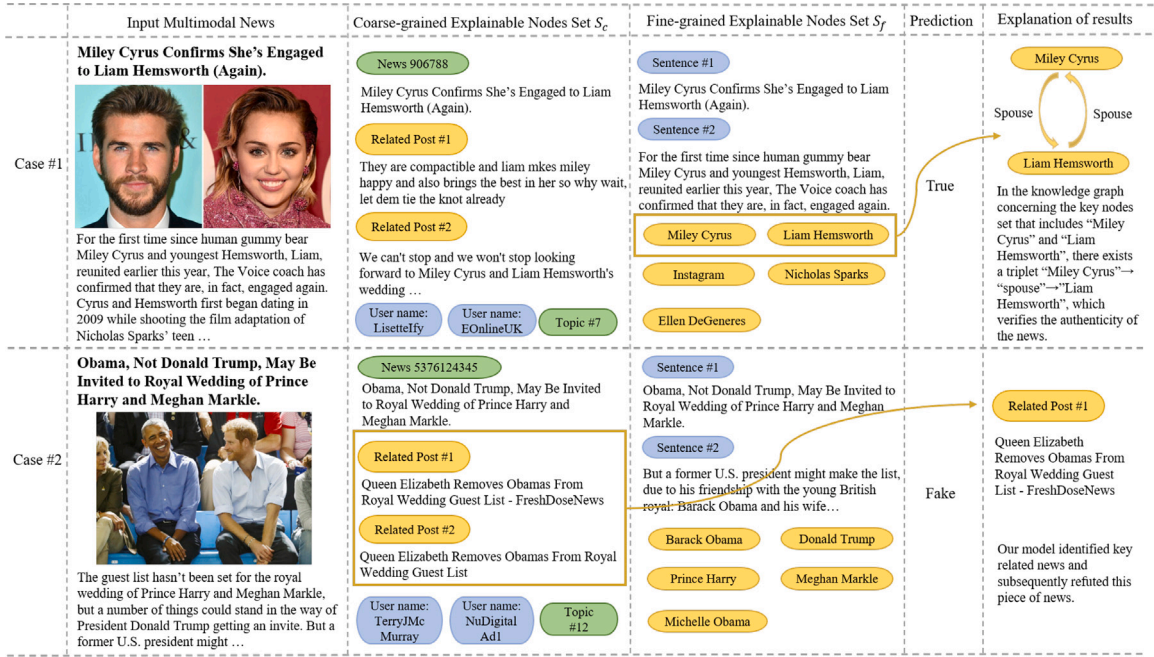


Fig. 5. Case studies on GossipCop and Politifact dataset.

(Again).’ In the second example, our model accurately extracted the related post ‘Queen Elizabeth Removes Obamas From Royal Wedding Guest List - FreshDoseNews.’ This news counters the claim made in ‘Obama, Not Donald Trump, May Be Invited to Royal Wedding of Prince Harry and Meghan Markle,’ and can serve as an explanation for categorizing it as fake. Consequently, MRE-FND effectively integrates social network, news content, and knowledge graph information, enhancing the model’s classification performance and providing explanations for the classification results using knowledge from various sources.

6.7. Discussion

In this section, we discuss whether the present study has achieved the research objectives set forth. **Firstly**, we developed a dual-layer heterogeneous graph model that encompasses multiple sources of data, including users, source news, related posts, and a knowledge graph. The experimental results demonstrate that our model effectively integrates information from diverse sources. Our model achieves state-of-the-art performance on the Politifact, Gossipcop and PHEME datasets. Ablation studies confirm that the model’s performance decreases in the absence of any source of information, validating our model’s capability to effectively utilize information from various sources. **Secondly**, we designed a dual graph neural network to extract and integrate valuable information from multiple sources. The results from ablation studies indicate that the absence of the dual graph neural network module leads to a significant decrease in the model’s performance on both the Politifact and Gossipcop datasets, thereby demonstrating the efficacy of our dual graph neural network design. **Thirdly**, we designed an explainable reasoning module to extract meaningful knowledge from complex information and generate classification explanations based on this. The explainability analysis experiments reveal that our model accurately extracts the key information necessary for detecting fake news. Ablation studies demonstrate a decline in model performance on the Politifact and Gossipcop datasets when the explainable reasoning module is absent, thereby affirming the effectiveness of this module.

7. Implication

In recent years, social media platforms have gained popularity as avenues for individuals to communicate messages unrestricted by geographical distance. However, the accessibility and efficiency of social media also facilitate the rapid dissemination of misinformation, potentially leading to widespread panic and social turmoil. The multimodal news, incorporating text and images, garners widespread attention due to its rich content. As the generation of increasingly nuanced fake news rises, traditional fake news detection methods relying solely on news content or partial external information prove insufficient. Furthermore, current fake news detection models lack explainability in their classification results, further diminishing the credibility of the models. Therefore, we propose the MRE-FND model for explainable fake news detection, which integrates diverse information sources and provides explanations for its outcomes. We performed comprehensive experiments on two datasets to showcase the efficacy of our MRE-FND approach. The experimental results indicate that our model achieved accuracy rates of 92.9%, 83.4% and 84.7% on the Politifact, Gossipcop and PHEME datasets, respectively, representing improvements of 2.0, 0.8 and 1.1 percentage points over the baseline.

Compared to previous methods of fake news detection, our model integrates not only the textual and visual information from the news itself but also external information, including user networks, related posts, topic information, and knowledge graphs. To the best of our knowledge, this represents the first multimodal framework for fake news detection, which integrates various information sources, including social networks, external news, news topics, and knowledge graphs, enabling comprehensive acquisition of pertinent information for fake news detection. Furthermore, our model is capable of generating interpretable node sets from both coarse-grained and fine-grained perspectives to provide multi-faceted explanations for classification outcomes. Our approach enhances both the accuracy and explainability of fake news detection, thereby offering new insights for future research.

8. Conclusion

With the development of social media, fake news has exhibited diversity and complexity. Traditional detection methods, relying solely on content and limited external information, are inadequate to address new challenges. Moreover, while different sources such as social networks and knowledge graphs offer more features, they also introduce more noise, and current methods lack explainability in fake news classification. Therefore, we propose a novel Multifaceted Reasoning Network for Explainable Fake News Detection (MRE-FND). Our model effectively integrates various external knowledge sources, selectively utilizing valuable information for fake news classification and generating explainable nodes. We constructed a dual heterogeneous network containing various types of news information, including a coarse-grained news social graph and a fine-grained news knowledge graph. This network integrates diverse source information such as original news, related posts, user information, thematic information, news sentences, and news entities. Subsequently, we design a dynamic heterogeneous graph neural network based on the Graph Information Bottleneck principle. It dynamically updates node representations while dynamically adding or removing nodes and edges. Utilizing the updated dual-heterogeneous graph, we employ explainable reasoning module to extract explainable nodes. In the end, we perform fake news classification based on the news knowledge subgraph, social subgraph, and explainable nodes. Extensive experiments are conducted to assess performance, with results illustrating the superiority of our MRE-FND model. Overall, our model effectively handles redundant information from multiple sources. The explanations for the classification results, based on essential knowledge, also make the model more persuasive. In future work, we will explore incorporating state-of-the-art large-scale model approaches to generate more readable classification explanations while accomplishing fake news detection.

CRedit authorship contribution statement

Lin Feng Han: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xiao Ming Zhang:** Writing – review & editing, Supervision, Funding acquisition, Formal analysis. **Zi Yi Zhou:** Writing – original draft, Validation, Data curation. **Yun Liu:** Writing – review & editing.

Data availability

Data will be made available on request.

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